Lab 2 Silver Equilibrium 3/10

1. Determine the Nerst Equation

From raw data:

$E_1 = 0.3037 \pm 0.0007131\ V$

$E_2 = 0.2495 \pm 0.0006730\ V$

$E_3 = -0.25711 \pm 0.0004560\ V$

$E_4 = -0.09268 \pm 0.0004496\ V$

$E_5 = -0.1121 \pm 0.0004502\ V$

$E_6 = -0.2225 \pm 0.0004663\ V$

From guide, the Nerst equation is given by:

$$E_{obs} = const. + 0.0592 \lg[Ag^+]$$

As the initial concentration is $[Ag^+]_0 = 1.00 \times 10^{-4} M$,

$$Const. = E_1 - 0.0592 \lg(1.00 \times 10^{-4})$$

$$= 0.3037 \pm 0.0007131\ V + 4 \cdot 0.0592$$

$$= 0.5405 \pm 0.0007131\ V$$

2. Calculation for AgCl

(1) Calculation of $[Ag^+]$ & Q_{sp}

Using Nerst equation:

$$E_{obs} = const. + 0.0592 \lg[Ag^+]$$

$$[Ag^+] = 10^{(E_{obs} - 0.5405)/0.0592}$$

Use E_2: $[Ag^+]_{AgCl} = 10^{(0.2495-0.5405)/0.0592}$ M

$= 10^{-0.291/0.0592}$ M

$\approx 1.215\times10^{-5}$ M

Assume $[Ag^+]=[Cl^-]$,

$Q_{sp}(AgCl) = [Ag^+]^2_{AgCl} = (10^{-0.291/0.0592})^2 \approx 1.475\times10^{-10}$

(2) Error Analysis I

Let $y = 10^{(E_{obs}-C)/0.0592}$,

① $\frac{\partial y}{\partial E_{obs}} = 10^{(E_{obs}-C)/0.0592}\cdot \ln(10)\cdot\frac{1}{0.0592} = y\frac{\ln 10}{0.0592}$

② $\frac{\partial y}{\partial C} = 10^{(E_{obs}-C)/0.0592}\cdot \ln(10)\left(-\frac{1}{0.0592}\right) = -y\frac{\ln 10}{0.0592}$

$$S_y = \sqrt{\left(y\frac{\ln 10}{0.0592}S_E\right)^2+\left(-y\frac{\ln 10}{0.0592}S_C\right)^2} = y\frac{\ln 10}{0.0592}\sqrt{S_E^2+S_C^2}$$

From previous calculations & experimental data:

$S_E = 0.0006730$ $S_C = 0.0007131$

$S_y = 1.215\times10^{-5}\cdot\frac{\ln 10}{0.0592}\cdot\sqrt{(6.730\times10^{-4})^2+(7.131\times10^{-4})^2} \approx 0.0463\times10^{-5}$

Thus, $[Ag^+]_{AgCl} = (1.215\pm0.0463)\times10^{-5}$ M

As $Q = C_{Ag^+}C_{Cl^-} = C^2_{Ag^+}$, $\frac{dQ}{dC} = 2C$

$S_Q = 2C\cdot S_C = 2\cdot1.215\times10^{-5}\cdot4.634\times10^{-7} \approx 0.113\times10^{-10}$

So, $Q(AgCl) = (1.475\pm0.113)\times10^{-10}$

(3) Calculation for $\gamma_\pm$

$n(NH_4NO_3) = 0.01M \cdot 0.010L = 1.00\times10^{-4} mol$

$V_{total} = 10mL + 100mL = 0.110L$

$[NH_4NO_3] = \frac{1.00\times10^{-4}}{0.110} M \approx 9.09\times10^{-4} M$

From lecture: $I = \frac{1}{2}\sum_i z_i^2 m_i$, $z_1 = z_2$ & $m_1 = m_2$ for NH_4NO_3

As $[NH_4^+] = [NO_3^-]$, assume $25^\circ C$, $A = 0.5092$, from lecture:

$$\log\gamma_\pm = -|z_+z_-|AI^{\frac{1}{2}}$$
$$= -0.5092 I^{\frac{1}{2}}$$
$$= -0.5092 \cdot \sqrt{\frac{1}{2}\cdot(1+1)\cdot\frac{1.00\times10^{-4}}{0.110}}$$
$$\approx -0.01535$$
$$\gamma_\pm = 10^{-0.01535} \approx 0.9653$$

(4) Calculation for K_{sp}

$K_{sp} = \gamma_\pm^2 Q_{sp} = 0.9653^2 \cdot (1.475\times10^{-10}) \approx 1.375\times10^{-10}$

(5) Error Analysis II

Since $K = \gamma_\pm^2 Q$, $S_K = \gamma_\pm^2 S_Q = 0.9653^2 \cdot 0.113\times10^{-10} \approx 0.105\times10^{-10}$

So, $K_{sp}(AgCl) = (1.375 \pm 0.105)\times10^{-10} M$

Literature: $K_{sp} = 1.77\times10^{-10}$

error rate $= \frac{|(1.375-1.77)\times10^{-10}|}{1.77\times10^{-10}} \times 100\% = \frac{0.395}{1.77}\times100\% \approx 22.32\%$

3. Calculation for AgI

(1) Calculation for $[Ag^+]$, $[I^-]$ & Q_{sp}

Use previously derived Nerst equation & E_6:

$$[Ag^+]_{AgI} = 10^{(-0.2225-0.5405)/0.0592} \approx 1.293\times10^{-13}\,M$$

$$n(I^-)_{total} = 0.010L \cdot 0.01M = 1.00\times10^{-4}\,mol$$

$$n(Ag^+)_{ini} = 0.010L \cdot 0.001M = 1.00\times10^{-5}\,mol$$

$$n(I^-)_{res} = 1.00\times10^{-4}mol - 1.00\times10^{-5}mol = 9.00\times10^{-5}mol$$

$$V_{final} = 100mL + 10mL + 30mL + 10mL = 0.150L$$

$$[I^-]_{res} = \frac{9.00\times10^{-5}}{0.150}M = 6.00\times10^{-4}M$$

$$Q_{sp} = [Ag^+]_{AgI}\cdot[I^-]_{res} = 10^{(-0.2225-0.5405)/0.0592}\cdot 6.00\times10^{-4} \approx 7.756\times10^{-13}$$

(2) Error Analysis I

From previous:

$$S(Ag^+) = [Ag^+]_{AgI}\frac{\ln10}{0.0592}\sqrt{S_{E_6}^2 + S_C^2}$$

$$= 10^{(-0.2225-0.5405)/0.0592}\cdot\frac{\ln10}{0.0592}\sqrt{(4.663\times10^{-4})^2 + (2.131\times10^{-4})^2}$$

$$\approx 0.043\times10^{-13}$$

So, $[Ag^+]_{AgI} = (1.293 \pm 0.043)\times10^{-13}\,M$

$$S_Q = [I^-]_{res}\cdot S(Ag^+) = 6.00\times10^{-4}\cdot 4.2838\times10^{-15} \approx 0.2570\times10^{-17}$$

So, $Q_{sp} = (7.756 \pm 0.2570)\times10^{-17}$

(3) Calculation for $\gamma_\pm$

$n_{total}(NH_4NO_3) = 1.00\times10^{-4}\,mol \cdot 4 = 4.00\times10^{-4}\,mol$

$[NH_4NO_3] = \frac{4.00\times10^{-4}}{0.150}\,M \approx 2.667\times10^{-3}\,M$

From previous, $\gamma_\pm = 10^{-0.509\cdot\sqrt{2.667\times10^{-3}}} \approx 0.9413$

(4) Calculation for K_{sp}

$K_{sp} = \gamma_\pm^2 Q_{sp} = (0.9413)^2\cdot(7.759\times10^{-17}) \approx 6.872\times10^{-17}$

(5) Error Analysis II

$S_K = \gamma_\pm^2 \cdot S_Q = 0.9413^2 \cdot 2.570\times10^{-18} \approx 2.277\times10^{-18}$

So, $K_{sp} = (6.872 \pm 0.2277)\times10^{-17}$

Literature: $K_{sp}(AgI) = 8.52\times10^{-17}$

error rate $= \frac{|(8.52-6.872)\times10^{-17}|}{8.52\times10^{-17}} \times 100\% \approx 19.34\%$

4. Formation Constant & K_f

(1) $[Ag^+]$, $[NH_3]$ & $[Ag(NH_3)^+]$

$[Ag^+]_3 = 10^{(E_3-C)/0.0592} = 10^{(-0.05711-0.5405)/0.0592} \approx 8.040\times10^{-11}\,M$

$S_{[Ag^+]_3} = [Ag^+]_3 \frac{\ln 10}{0.0592}\sqrt{S_{E_3}^2+S_C^2} \approx 0.2647\times10^{-11}\,M$

$[NH_3]_3 = \frac{10.200}{120}\,M \approx 0.1667\,M$

$[Ag(NH_3)^+]_3 = 1.000\times10^{-4}\cdot\frac{100}{120} \approx 8.333\times10^{-5}\,M$

So, $[Ag^+]_3 = (8.040 \pm 0.2647)\times10^{-11}\,M$

$$[Ag^+]_4 = 10^{(E_4-C)/0.0592} = 10^{(-0.2268-0.5405)/0.0592} \approx 2.016 \times 10^{-11} M$$

$$S_{[Ag^+]_4} = [Ag^+]_4 \frac{\ln 10}{0.0592} \sqrt{S_{E_4}^2 + S_C^2} \approx 0.066 \times 10^{-11} M$$

$$[NH_3]_4 = \frac{20.2.00}{130} M \approx 0.3077 M$$

$$[Ag(NH_3)^+]_4 = 1.000 \times 10^{-4} \cdot \frac{100}{130} \approx 7.692 \times 10^{-5} M$$

So $[Ag^+]_4 = (2.016 \pm 0.066) \times 10^{-11} M$

$$[Ag^+]_5 = 10^{(E_5-C)/0.0592} = 10^{(-0.1121-0.5405)/0.0592} \approx 9.470 \times 10^{-12} M$$

$$S_{[Ag^+]_5} = [Ag^+]_5 \frac{\ln 10}{0.0592} \sqrt{S_{E_5}^2 + S_C^2} \approx 0.3106 \times 10^{-11} M$$

$$[NH_3]_5 = \frac{30.2.00}{140} M \approx 0.4286 M$$

$$[Ag(NH_3)^+]_5 = 1.000 \times 10^{-4} \cdot \frac{100}{140} \approx 7.143 \times 10^{-5} M$$

So $[Ag^+]_5 = (9.470 \pm 0.3106) \times 10^{-11} M$

(2) Linear Regression

Point 1: $\log(NH_3) = \log(0.1667) \approx -0.7782$

$$\log\left(\frac{[Ag(NH_3)^+]}{[Ag^+]}\right) = \log\left(\frac{8.333 \times 10^{-5}}{8.040 \times 10^{-11}}\right) \approx 6.016$$

Let $y = \log\left(\frac{[Ag(NH_3)^+]}{[Ag^+]}\right)$, $\frac{\partial y}{\partial [Ag^+]} = -\frac{1}{[Ag^+]\ln 10}$

$$S_y = \frac{S_{[Ag^+]}}{[Ag^+]\ln 10} = \frac{2.647 \times 10^{-12}}{8.040 \times 10^{-11} \ln 10} \approx 0.01430$$

Point 2: $\log(NH_3) = \log(0.3077) \approx -0.5119$

$$\log\left(\frac{[Ag(NH_3)^+]}{[Ag^+]}\right) = \log\left(\frac{7.692 \times 10^{-5}}{2.016 \times 10^{-11}}\right) \approx 6.582$$

$$S_y = \frac{S_{[Ag^+]}}{[Ag^+]\ln 10} = \frac{0.066 \times 10^{-11}}{2.016 \times 10^{-11} \ln 10} \approx 0.01422$$

Point 3: $\log(NH_3) = \log(0.4286) \approx -0.3680$

$$\log\left(\frac{[Ag(NH_3)^+]}{[Ag^+]}\right) = \log\left(\frac{7.143\times10^{-5}}{9.470\times10^{-11}}\right) \approx 6.878$$

$$S_\gamma = \frac{S_{[Ag^+]}}{[Ag^+]\ln 10} = \frac{0.3106\times10^{-11}}{9.470\times10^{-11}\ln 10} \approx 0.01424$$

From linear regression,

$m = 2.1045 \quad S_m = 0.0485$

$b = 7.6547 \quad S_b = 0.0280$

So, $p = 2.104 \pm 0.04850$. Take $p = 2$

As $\log K_f = b$, $K_f = 10^{7.6547} \approx 4.515\times10^7$

Since $K_f = 10^b$, $\frac{dK_f}{db} = \ln 10 \cdot 10^b = \ln 10 K_f$

$$S_{K_f} = \ln 10 \cdot K_f \cdot S_b = \ln 10 \cdot 4.515\times10^7 \cdot 0.0280 \approx 2.911\times10^6$$

So, $K_f = (4.515 \pm 0.2911)\times10^7$

(3) Calculation for K_f from p

As $p=2$, $K_f = \frac{[Ag(NH_3)_2^+]}{[Ag^+][NH_3]^2}$

Point 1: $K_{f_1} = \frac{8.333\times10^{-5}}{(8.040\times10^{-11})(0.1667)^2} \approx 3.732\times10^7$

$$S_{K_{f_1}} = K_f \frac{S_{[Ag^+]_3}}{[Ag^+]_3} = 3.732\times10^7 \cdot \frac{2.647\times10^{-12}}{8.040\times10^{-11}} \approx 1.229\times10^6$$

Point 2: $K_{f_2} = \frac{7.692\times10^{-5}}{(2.016\times10^{-11})(0.3077)^2} \approx 4.031\times10^7$

$$S_{K_{f_2}} = 4.031\times10^7 \cdot \frac{0.66\times10^{-12}}{2.016\times10^{-11}} \approx 1.328\times10^6$$

Point 3: $K_{f_3} = \frac{2.143 \times 10^{-5}}{(9.470 \times 10^{-11})(0.4286)^2} \approx 4.107 \times 10^7$

$$S_{K_{f_3}} = 4.107 \times 10^7 \cdot \frac{3.186 \times 10^{-12}}{9.470 \times 10^{-11}} \approx 1.347 \times 10^6$$

$$\overline{K_f} = \frac{K_{f_1} + K_{f_2} + K_{f_3}}{3} \approx 3.956 \times 10^7$$

$$S_{\overline{K_f}} = \sqrt{\frac{\sum_i (K_{f_i} - \overline{K_f})}{2}} \approx 1.984 \times 10^6$$

From table, $t = 4.303$ for 95% interval

$$\lambda_{K_f} = t \frac{S_{K_f}}{\sqrt{n}} = 4.303 \frac{1.984 \times 10^6}{\sqrt{3}} \approx 0.4929 \times 10^7$$

So, $K_f = (3.956 \pm 0.4929) \times 10^7$

From literature $K_f = 1.6 \times 10^7$,

error rate 1 $= \frac{|4.515 - 1.6| \times 10^7}{1.6 \times 10^7} \times 100\% \approx 182.2\%$

error rate 2 $= \frac{|3.956 - 1.6| \times 10^7}{1.6 \times 10^7} \times 100\% \approx 147.3\%$